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Q1) Determine whether the graphs G_1 and G_2 are isomorphic. Prove your answer.

a)

Number of verticies in $G_1 = 5$

" " " " $G_2 = 5$

Number " edges " $G_1 = 8$

" " " " $G_2 = 8$

Degree =>

G_1	G_2
deg(a) = 8	deg(1) = 3
deg(b) = 4	deg(2) = 4
deg(c) = 2	deg(3) = 3
deg(d) = 4	deg(4) = 3
deg(e) = 8	deg(5) = 3

→ Degree of vertices are not similar

$G_1 \ncong G_2$

G_1 is not isomorphic to G_2.

b) Number of verticies in $G_1 = 7$

" " " $G_2 = 7$

" " edges " $G_1 = 11$

" " edges " $G_2 = 11$

G_1	G_2
deg(a) = 3	deg(1) = 3
deg(b) = 3	deg(2) = 3
deg(c) = 3	deg(3) = 3
deg(d) = 3	deg(4) = 3
deg(e) = 4	deg(5) = 3
deg(f) = 3	deg(6) = 4
deg(g) = 3	deg(7) = 3

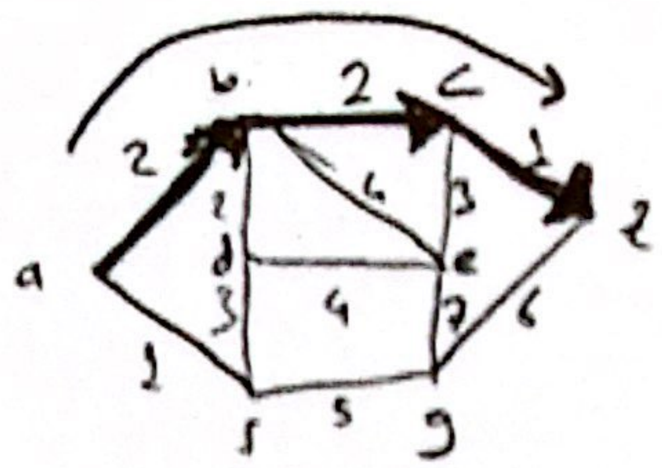
→ Degree of similar

$G_1 \cong G_2$

G_1 is isomorphic to G_2

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Q2)



a)

a $\xrightarrow{2}$ b $\xrightarrow{2}$ c $\xrightarrow{1}$ z

shortest path

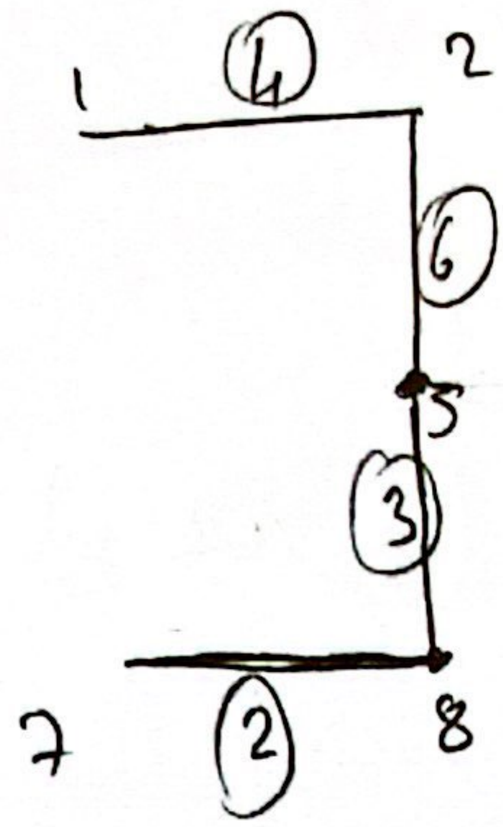
abcz

b) 5

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Q3)

a) steps! →



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Q4)

a)

Input			Output	
x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

b)

$$C = (x \text{ AND } y) \text{ OR } (z \text{ AND } (x \text{ OR } y))$$

(S) sum = $(x \text{ XOR } y) \text{ XOR } z$

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